

AI Ethics in the LAC Region

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This module walks the audience through the state of the AI and ethics ecosystem in Latin America and the Caribbean. People tend to focus on the United States, the European Union and China when thinking about artificial intelligence, ignoring innovative developments in AI technologies for the public sector emerging from the LAC region. Cristina Pombo discusses numerous efforts Latin American and Caribbean countries are making to promote the responsible use of AI. She also shares a few examples of creative uses of AI technologies to address challenges unique to the LAC region. The module concludes with an overview of the tools developed by the Inter-American Development Bank's fAIr LAC initiative that help bring ethical AI principles into practice.

Lecture Transcript

0:02 Hello, my name is Christina Pombo. And I work at the Inter-American Development Bank Group, one of the most important sources of financing and technical support for economic, social and institutional development in the Latin American and the Caribbean region. Specifically, my focus is in supporting the advancement of the ethical use of technology, in particular artificial intelligence, in the social services. Since the responsible use of AI and its potential to improve the social well being began to be discussed, many governments in the region have turned their attention towards solving large scale social problems such as education, poverty, and inequality among others. To the extent that artificial intelligence gradually becomes known as a technology that is accessible and has a general purpose in daily life, its impact will be graded in terms of its broader application to all aspects of human existence. Artificial intelligence applications are diverse, and their growth is noticeable in different spheres of life, where patterns may be detected from big volumes of data and complex models, as well as in the availability of independent systems that can improve decision making and generate more egalitarian and efficient policies. Therefore, in collaboration with partners and strategic allies, the Inter-American Development Bank, or IDB, created the fAIr LAC initiative to promote the responsible adoption of artificial intelligence and decision support systems to improve social services delivery and create development opportunities to reduce social inequality. Today, we are going to explore if Latin America and the Caribbean is a relevant player in the AI and ethics ecosystem. People tend to think that artificial intelligence is far away from this region, that things are only happening in China, the European Union or the United States. But that's not entirely true. This module will walk you through the different efforts Latin American and Caribbean countries are doing to adopt a responsible use of AI and some practical tools developed by the Inter-American Development Bank to support those efforts in going from principles to practice.

2:31 Latin America and the Caribbean covers an extensive region extending from the Bahamas and Mexico to Argentina and Chile. The region has a population of over 750 million people as of 2019, and in 2020, the GDP

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was \$4.7 trillion. This is slightly over 20% of the US GDP, which is a great opportunity. The region has long been the victim of low productivity and the COVID-19 pandemic is predictably making matters worse. Now, artificial intelligence is a chance for the region's economics, economists, to leap for the greater innovation and economic progress. Research suggests that AI will add a full percentage point of GDP to five of South America's biggest economies - Argentina, Brazil, Chile, Colombia and Peru by 2035. To give you just one example of how machine learning is bringing unique Latin American solutions to unique Latin American challenges, think about how e-commerce firms have faced a conundrum in Latin America. How can they deliver packages in a region where 25th percent of urban populations live in informal scattered neighborhoods with no addresses? Data is key. A logistic startup from Peru, which partnered with Arequipa's University of San Pablo to build an artificial intelligence robot to generate new postal maps across the country. The company has now expanded to Argentina, Mexico and Chile introducing remote communities and city outskirts to online deliveries. Artificial Intelligence and its correlated technologies could prove a major boon to the region's public and private sectors. In turn, its policymakers and business leaders must better prepare to take full advantage while warding off potential downsides. In a region that suffers from endemic corruption, pervasive violence, weak institutions and challenging socio economic conditions, governments, policymakers and organizations can use artificial intelligence to tackle critical issues in the region, including food security, health, education, natural resources, and unemployment.

5:04 In order why, for example, artificial intelligence is being used to reduce the number of students drop from high school. The objective is to generate a predictive model, an early warning system for educational disengagement in secondary education based on information from students and the conditions of the educational offer. This would provide information to different actors of the educational system and families to inform the concrete actions that are carried out at the territorial and school level to protect the educational pathways of the students. In Brazil, the University of São Paulo is developing machine learning technology that will rapidly assess the likelihood that patients have Dengue fever, Zika or Chikungunya when they arrive at a medical center. As it has been discussed in other models, the use of artificial intelligence systems in decision making processes has associated risk due to the potential direct or indirect impact that such decisions may have on the lives of the users of affected populations. Being developed by humans and fed with data, AI can incorporate implicit biases that comes from the judgment of the people who design the tool, or that are the result of incomplete or low quality information. These risks can be anticipated and mitigated by involving people who constantly review the results of the systems and establishing clear protocols at each stage of the lifecycle of any AI system. This approach makes it easy to organize the discussion and identify the specific risks associated with each stage. We're given the iterative nature of AI, a linear approach is not appropriate. In the public sector, this approach facilitates the connection with the public policy cycle, and presents technology, particular AI, as an available tool to develop the proposed public policy alternatives based on the context and feasibility of the solution. In addition, it helps to put the need or problem at the center, avoiding a techno-solutionist approach. When the government chooses AI as the appropriate instruments that will respond to the prioritized problem, the risk mitigation analysis begins through the following stages of the AI lifecycle: conceptualization and design - understanding the key points that decision makers need to be clear about before starting a project; data collection and processing - it includes the data cleaning and processing process, as well as the identification of deficiencies and biases that could put the development of the model at risk. Third, model development and validation - understand the key concepts to follow to have robust and validated AI systems. Four use and monitoring - it includes the evaluation of the tool

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once the implementation has begun. And finally, accountability responds to the need to provide information and transparency to promote understanding of AI by all citizens. At each of these stages, there are critical factors and decisions that help determine whether the AI system's ethical, fair and accountable, and help guarantee the AI solution effectively supports a public policy strategy. Recent years have seen the proliferation of frameworks to guide the ethical use of artificial intelligence systems. Such frameworks have been focused on the formulation, development or implementation of AI systems, but often also aimed at regulating the use of AI by governments. The Berkman Klein Center for Internet and Society carried out a systematic and comparative mapping, where it analyzed 36 sets of ethical principles and current guidelines, where despite their diversity in terms of audience, composition, scope and depth, each one of them has the same purpose of present a vision for AI governance. The document is written by different actors between governments and intergovernmental organizations, companies, professional associations and civil society groups, coinciding highlighting eight common themes: privacy, accountability, protection and security, transparency and explicability, equity and non discrimination, human control of technology professional, responsible and promoting of human values.

10:13 Most of these documents are frameworks that seek to systematize non mandatory recommendations of ideal but not mandatory adoption. An example of a reference framework is the Recommendation on the Ethics of Artificial Intelligence of the United Nations Educational, Scientific and Cultural Organization (UNESCO) recently approved. This framework has one of its main objectives to address various concerns regarding the use of systems based on artificial intelligence to achieve an inclusive and pluralistic instrument that guarantees the development and use of AI systems in accordance with the principles of inclusion, diversity and transparency. In the case of Latin America and the Caribbean, the principles on artificial intelligence of the Organisation for Economic Cooperation and Development (OECD) have been particularly relevant given that in May 2019, Argentina, Brazil, Colombia, Costa Rica and Peru, together with the 36 member countries of the OECD, signed the principles in Paris. On the other hand, there are regulatory frameworks for the responsible implementation of artificial intelligence systems, which are recognized by national or international law and human rights. They have a standard with their ability to enforce compliance and have globally recognized validity, such as the White House strategy or the European Framework. The governments of the region are developing artificial intelligence strategies. In an uneven way, yes, because different approaches and different speeds are being seen. However, there is an acceleration and intensification of the debate on public policies and government strategies in recent years. Argentina, Brazil, Chile, Colombia, Mexico, Peru and Uruguay have adopted national strategies in AI or are in the process of doing so. Most countries that do not have an AI strategy have a more general one for digital government or a related agenda or program. Some countries have shown relative strength in certain components of an AI strategy that could benefit other countries if working in close collaboration. For example: Argentina in experimentation; Brazil in interoperability; Chile in understanding the users needs; Colombia in ensuring an ethical and trustworthy approach; and Panama in infrastructure; Uruguay in underlying data strategy, etc. To define and successfully implement the strategy, governments must have certain critical capabilities. Those capabilities will enable proper use of AI complying with the standards in ethical principles: equity and mitigation of biases, transparency and explainability, safety and responsibility, inclusive and focused approaches in the user, and spaces of experimentations with AI.

13:13 Not all countries have explicitly included action to acquire these capacities in their national programs, and that is a challenge. Colombia and Uruguay stand out but other countries have also expressed this intention to

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create this capacity in its national strategy or policies. In Latin American and the Caribbean, there are some regulations in place mainly related to data protection. Brazil has a general law of protection of personal data, which entered into force on 2020. It is a comprehensive data protection law that covers the activities of data controllers and processors, and creates modern requirements on the treatment of people's personal information. The regulation is one of the most recent in the Latin American region and contains some novel rules based on the GDPR related to the use of personal data in artificial intelligence solutions. Argentina has a regulatory framework for the protection of personal data, the consumer and the right to public information, and since the 20s, they have a loan personal data protection, which offers free access to data generated or processed by artificial intelligence systems. Uruguay has protection of personal data and access to public information law since 2008. The law provides a series of definitions and principles that allow their integration in the event of gaps. In the case of applications related to artificial intelligence, there are no specific regulations that refer to it without prejudice, to which the government has allocated substantial resources to obtain this type of technology in the public sector. Ecuador has a guide for the processing of personal data in the administration central public of the country. Although it is not at the same legal level, as the legislation of data protection requires that the processing of personal data, but public entities may not cause discrimination of any kind. Some countries have designed their own principles for AI as a positive step towards achieving an environment and a culture of trust regarding the development of use of this technology. In the last few years, effort has been accelerated to ensure reliable and ethical AI system and policies specific to each country, with the following advances. In 2018, Mexico published 14 principles for the development and use of AI. In 2019, Uruguay included nine general principles as part of its AI strategy to guide digital transformation in government entities and provide a framework for the use of this technology in the public sphere. In 2020, Colombia and Chile published drafts for consultation on their AI strategy. Colombia developed the ethical framework for artificial intelligence and a plan to hold roundtables of experts to provide feedback and develop a final version. Chile includes its cross cutting principle as part of its AI policy. Brazil committed in its 2021 national AI strategy to formulate ethical principles for the design and implementation of systems that use this technology. While ethics was a strong focus on the Brazilian strategy, the specific scope and content of its ethical principle is not yet established. And Costa Rica started the process to create a roadmap for the establishment of the guiding framework for the responsible and ethical use of artificial intelligence. The principles developed emphasize local priorities and the specific context of each country. For example, Chile includes sustainable growth with environmental protection and multidisciplinary as the default approach to AI and the global reach and impact of these systems. Colombia incorporates a measure to protect the rights of children and adolescents. For Mexico, measuring impact is essential to ensure that AI systems comply with its purposes. Uruguay emphasizes that technological development in this area must have a purpose to complement and add value to human activities. And Costa Rica is putting a focus on talent as part of the climate investment strategy. The development of regulatory frameworks for AI is at an early stage in the region, presenting an opportunity for this to be discussed and shared with all the actors involved. Progress in this direction is crucial to define a regulation focused on ethics, security, justice and transparency, which strengthens the trust of citizens in this technology. The progress made will also require frequent public reviews and scrutiny to maintain relevance, applicability and effectiveness.

18:23 Let's talk a little bit about the tools for the implementation of ethical principles in the region. The guarantee of equity and the mitigation of biases is based on the capacity of the countries to establish safeguards against prejudice and injustice that prevents AI to reinforce certain forms of discrimination that may exist, such as racism

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or sexism. Some countries in the region stand out for their commitment to equity, non discrimination and harm prevention principles. As part of its ethical framework, Columbia is developing an algorithm registration system to perform ethical evaluations on AI projects as a tool for transparency and explicability in the public sector. Columbia, Mexico and Uruguay provide a clear role for humans to maintain control of AI systems and make corrections when necessary. Uruguay establishes a social objective to protect the general interest and guarantee the inclusion and equity and more specifically, it states that specific work must be done to do reduce the possibility of unwanted biases in the data and model used that may negatively impact the people favoring discriminatory practices. And Chile calls for non discrimination or detriment to any group and emphasize to the consideration of childhood, adolescence and the gender perspective. Unlike other countries, the Chilean principles do not explicitly mention privacy. Additionally, countries of the region are advancing another initiative to establish safeguards against prejudice and injustice showing great potential. Argentina's artificial intelligence strategy recognizes the risk of bias in AI systems as part of the diagnosis of the transversal axes, ethics and regulation, although it does not include the specific measures. Brazil's nation's AI strategy includes action items to develop techniques for an identification and mitigation of algorithm bias and to ensure data quality for systems training. It is also intended to direct funds toward projects that support fairness, non discrimination and diversity in AI development teams. It is committed to developing approaches to reinforce the role of human beings to mitigate risks. Chile's AI policy proposes the creation of new institutions capable of establishing precautionary actions and promote research against bias and injustice - a cross cutting axes as it considers the reduction of biases related to gender, the bias data production and development teams with little diversity. And Uruguay has launched two relevant instruments to confront bias and injustice: the framework for data quality management and the impact study model algorithm specifically aimed at AI. In Mexico, fAIr LAC Jalisco is a hub of the IDB's fAIr LAC regional initiative, where the government of Jalisco together with the Tecnológico de Monterrey and the BID Lab team up to advance an ambitious agenda for development of AI in the state. As part of the governance of AI, Jalisco created an Ethics Council. This council is a body whose function is to help maximize the opportunities and benefits of the use of autonomous or intelligent systems for social good. The governments of Latin America have the challenge to develop a specific controls and frameworks and evolving guidance mechanisms to ensure AI implementations that are consistent with their principles and rules. As a regional level, the IDB has been developing resources to generate awareness in Spanish about the potential benefits, but also the great risk this technology could have. These resources included not only general documents, but also specific tools to bring those principles into practice. It is not automatic to move from principles to observable characteristics, it is necessary to identify other levels of analysis. Based on the approach proposed by the AI Ethics Impact Group, the idea is to go through the definition of criteria indicators until arriving at observable characteristics. Therefore, we in fAIr LAC are developing different tools that seek to facilitate this landing of ethical principles. First, an ethical self evaluation. This tool seeks to review the main areas that should be considered through the project to mitigate the level of ethical risk associated with AI, particularly in social programs, programs where the sensitivity of the information is higher in comparison to other types of projects.

23:34 In the end, we seek to have an evaluation of the project's risk level from an ethical perspective before its implementation. And we have an ethical AI assessment for actors of the entrepreneurial ecosystem. This tool for entrepreneurs allows an analysis of the technological solutions based on AI and data management. Second, we have a manual for design and implementation of AI projects. Its objective is to go along with the formulation and execution of an AI project. It is a practical guide on the different instances, actors and ethical risks that decision

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makers and technical teams must consider when defining the AI solution's scope. And the technical manual of AI lifecycle. It is a guide for technical teams with recommendation and good practices to avoid unexpected results in decision makers objectives. The manual covers all the ethical challenges that are commonly found in any application of this technology. Also, we have a massive open online course - how to make responsible use of artificial intelligence in the public sector. This MOOC addresses the concept, principle, challenges and opportunities of the ethical and responsible use of artificial intelligence for the public sector. It presents tools to guarantee minimum standards as well to strengthen the quality of data and AI models from their design to their implementation and monitoring. It is important to highlight that the tools follow a living process of iteration and calibration through practice and must always have the individual at the center to ensure that we are indeed promoting the ethics use of AI. Finally, I want to give you some brief examples on how some countries are applying these tools in their AI projects. In Mexico, in Jalisco, diabetic retinopathy is a chronic and progressive disease that has a prevalence of 31.5%. In some industrialized countries, it has become a leading cause of partial vision loss and blindness in adults. In the absence of symptoms at the beginning of the disease, a dialogical examination is crucial at least every six months. Early detection facilitates treatment to reduce vision loss and prevent blindness. We're implementing a screening program at the first level of attention, where patients would be able to have an eye test, an eye image, and then the system automatically transfer their photographs to the diabetic retinopathy validation department. This department is assisted by certified grades and the AI system, which help determine the patient's level of risk to transfer those cases with an advanced DR level to a second level hospital for early diagnosis and timely treatment. As of now, more than 200 patients in three health centers in Guadalajara have been benefited and this screening program will be one of the public policy tools for DR prevention at the state level. The goal is that the Secretaría de Salud will provide the needed resources to implement it at its state level. In Argentina, inspired by the COVID-19 challenges and seeking to contain the risk of contagion of viruses and diseases in areas with large crowds, Distance2's technology platform was developed by the IDB. It takes advantage of video cameras already installed in cities. It uses modern AI algorithms to detect people and estimate the distance between them. Distance2 can generate alarms according to the parameters that the city determined for example, the percentage of people within two meters of distance in each unit of time, or deliver data sets that record the daily response of citizens to the measure provided. No private information or individual is used, since no particular trait or unique characteristics are analyzed. Only people are detected anonymously, and the distance between them is estimated.

28:05 In Ecuador, is working on including the preferences of families in the admission process to educational institutions. Currently, the system is centralized and based solely on the distance from homes to school. Some families not satisfied with their assignment request transfers, which represents 25% in Costa Rica and in the coastal zone and 18% in the Sierra-Amazonia zone. This transfer requests consume government resources, so this proposal is expected to reduce them. The country has now a platform that provides information on the educational offer available and families are able to select their preference from a prioritized list and incorporate some functionalities that use AI. For example, that the platform warns applicant families about a potential congestion in certain schools based on simulations that use data from previous years, or that the platform recommends a school based on the search and the interest declared by families during the school selection process. It is crucially important that governments develop these systems in ethical and responsible ways so that the biases of today are not baked into the technologies of tomorrow. Policymakers need to have courage and be encouraged to ask questions about how the technology is made and used, and then create robust and gender and diversity inclusive

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AI principles, guidelines and codes of ethics and ensure the correct implementations. As we have seen, Latin American countries and the Caribbean countries are well in the path to develop ethical and trustworthy AI systems. But a lot of work is yet to be done. You can find out more information about the cases and resources presented today in the fAIr LAC website and read, watch and listen all the resources we have. Thank you very much.